

Stony Brook Institute at Anhui University

2024 Spring Semester

Probability Theory Midterm Exam

(Closed-book Exam Duration: 120 minutes)

Section	I	II	III	IV	Total score
Score					
Grader					

I. Blank Fillings (5 items, 3 points each)

Score	
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1. Assume that the events E and F are mutually exclusive and $P(E)=0.6$, $P(F)=0.5$, then $P(EF^c)=$ _____.

2. Suppose that $P(A)=\frac{1}{4}$, $P(B|A)=\frac{1}{3}$ and $P(A|B)=\frac{1}{2}$, then $P(A\cup B)=$ _____.

3. Suppose that the probability mass function of a random variable X is $P(X=l)=a\cdot\left(\frac{3}{5}\right)^l$ ($l=0,1,2,\cdots$), then $a=$ _____.

4. Assume that a random variable $X\sim U(0,1)$ and let $Y=\frac{1}{\sqrt{\pi}}X^2$. Then the expected value of Y is _____.

5. Let a random variable $X\sim N(2,\sigma^2)$, $P(0<X<4)=0.3$, then $P(X<0)=$ _____.

II. Choices (5 items, 3 points each)

Score	
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6. Let two events A and B be mutually exclusive, $P(A)\neq 0$, $P(B)\neq 0$, then the right conclusion is ().

- A. A^c and B^c are mutually exclusive
- B. $P(B|A)>0$
- C. $P(AB)=P(A)P(B)$
- D. $P(A-B)=P(A)$

7. For any two events A and B , one has that ().

- A. $P(AB)\leq P(A)P(B)$
- B. $P(AB)\geq P(A)P(B)$

C. $P(AB) \leq \frac{P(A)+P(B)}{2}$ D. $P(AB) \geq \frac{P(A)+P(B)}{2}$

8. One person continues shooting at a goal with the success probability p in every shot. The probability that he succeeds for 2th time at the 5th shooting is ().

A. $5p^2(1-p)^3$ B. $5p^2(1-p)^2$ C. $4p^2(1-p)^3$ D. $2p^2(1-p)^3$

9. Let two events A and B be independent and $P(B)=0.5$, $P(AB^c)=0.3$, then $P(BA^c) = ()$.

A. 0.1 B. 0.2 C. 0.3 D. 0.4

10. Suppose that a random variable X follows $b(n, p)$, and $EX = 3.6$, $\text{Var}X = 2.16$, then the values of n, p are ().

A. $n = 6, p = 0.6$ B. $n = 60, p = 0.06$ C. $n = 9, p = 0.4$ D. $n = 4, p = 0.9$

III. Calculations (5 items, total 48 points)

Score	
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11. (8 points) From a group of 8 women and 6 men, a committee consisting of 3 men and 3 women is to be formed. **How many** different committees are possible if

(a) (4 points) 2 of the men refuse to serve together?

(b) (4 points) 1 man and 1 woman refuse to serve together?

12. (9 points) Suppose that a random variable X taking on one of the values -1, 0, 1 and 2 with respective probabilities $\frac{1}{2c}$, $\frac{3}{4c}$, $\frac{5}{8c}$ and $\frac{7}{16c}$. Calculate:

(a) (2 points) the constant c ;

(b) (3 points) $P(X < 1.5)$;

(c) (4 points) the expected value EX .

13. (10 points) A and B will take the same 10-question examination. Each question will be answered correctly by A with probability 0.7, independently of her results on other questions. Each question will be answered correctly by B with probability 0.4, independently both of her results on the other questions and on the performance of A.

Calculate:

(a) (5 points) Find the expected number of questions that are answered correctly by both A and B.

(b) (5 points) Find the variance of the number of questions that are answered correctly by either A or B.

14. (12 points) A worker has asked her supervisor for a letter of recommendation for a new job. She estimates that there is an 80 percent chance that she will get the job if she receives a strong recommendation, a 40 percent chance if she receives a moderately good recommendation, and a 10 percent chance if she receives a weak recommendation. She further estimates that the probabilities that the recommendation will be strong, moderate, and weak are 0.7, 0.2, and 0.1, respectively.

Calculate:

(a) (7 points) What is the probability that she will receive the new job offer?

(b) (5 points) Given that she does receive the new offer, what is the probability that that she received a strong recommendation?

15.(9 points) The density function of X is given by

$$f(x) = \begin{cases} a + bx^2, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

If $EX = 3/5$, **Calculate:**

(a) (2 points) the constants a and b ;

(b) (2 points) $P(X < 0.5)$;

(c) (5 points) the cumulative distribution function $F(x)$ of X .

IV. Proofs (4 items, total 22 points)

Score	
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16. (5 points) If $A \subset B$ and $P(A) = 1$, then $P(B) = 1$.

17. (5 points) Prove: $P(AB^c) = P(A) - P(AB)$.

18. (5 points) Let $F(x)$ be the distribution function of a random variable X . Prove: if $a < b$, then $F(a) \leq F(b)$.

19. (7 points) Let $X \sim P(\lambda)$. Prove $EX = \lambda$.